Project Evaluation Results

Datamites Project Feedback Report

Team	PTID-AIE-JUL-26-11142	**Project**	PRCP-1009
Course	AIE	**Overall Grade**	A
Topic	Cellphone Price Range		

Dear Team Member,

Please find below the detailed evaluation feedback for your project submission. Review each section carefully and use the improvement suggestions to guide your next steps.

EDA

Present Implementation

The submission performs a thorough initial check of the dataset, including shape, data types, and missing values. Visualizations include a count plot for the target variable and distribution histograms for key numerical features. Outliers are identified using boxplots and quantified via the Interquartile Range (IQR) method. A correlation heatmap is used to identify relationships between features and the target variable.

Scope for Improvement

While the EDA is comprehensive, the observations are largely descriptive. Connecting these observations to specific business hypotheses or explaining why certain features (e.g., RAM) show higher correlation would add analytical depth. Additionally, creating bivariate plots (e.g., scatter plots or box plots categorized by price range) would better illustrate the relationship between features and the target variable.

Preprocessing

Present Implementation

The submission implements a clean workflow for data preparation. It correctly splits the data into training and testing sets using stratification, which is excellent for maintaining class balance. Feature scaling is appropriately applied to algorithms that are sensitive to scale (Logistic Regression, KNN, SVM), while raw data is used for tree-based models, demonstrating a solid understanding of model requirements.

Scope for Improvement

The submission could benefit from justifying why certain features might need transformation or encoding if categorical features were present (though this dataset is primarily numerical). Explicitly confirming that no data leakage occurred during the scaling process for the entire dataset would strengthen the documentation.

Model Building

Present Implementation

The project implements a wide array of models (Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Gradient Boosting). It follows a systematic approach of training, evaluating, and comparing models using multiple metrics (Accuracy, Precision, Recall, F1-Score). Hyperparameter tuning using GridSearchCV is performed for the Logistic Regression model. The code for comparing model performance using a DataFrame and visualization is well-structured.

Scope for Improvement

Hyperparameter tuning is limited to the Logistic Regression model. Implementing tuning for the ensemble methods (Random Forest and Gradient Boosting) would likely improve their performance, potentially changing the conclusion regarding the best-performing model. Adding cross-validation scores during the initial model comparison would provide a more robust assessment than relying solely on a single train-test split.

Conclusion

Present Implementation

The conclusion clearly identifies the best-performing model and summarizes the findings effectively. It links the results back to the project objectives and provides actionable business insights based on the feature importance analysis. Limitations and future improvements are acknowledged, showing a holistic view of the project lifecycle.

Scope for Improvement

The conclusion asserts that Logistic Regression is superior to ensemble models. It would be beneficial to reflect on whether this is due to the lack of hyperparameter tuning on the ensemble models or if it represents a genuine finding regarding the data's linear separability.

Overall Feedback

Present Implementation

The project is well-structured, logical, and follows industry-standard practices for data science workflows. The submission demonstrates a strong command of Python libraries and machine learning fundamentals. The inclusion of feature importance analysis and a final "model testing" section on new data adds significant practical value to the project.

Scope for Improvement

The primary area for growth is in the depth of experimentation; applying hyperparameter tuning to high-performing ensemble models rather than just the baseline linear model would provide a more rigorous comparison.

Best regards,
LMT